# Cross-Lingual Information Extraction and Machine Translation System

## PART – 1

Task A: Multilingual Translation & Information Extraction Pipeline (8 Marks)

**Objective:**

Develop an integrated NLP pipeline that translates input text between English and an Indic language (e.g., Hindi, Tamil, Marathi, Bengali) while extracting key structured information—including Named Entities, Relations, and Temporal Events—from the text.

1. Web Interface (4 Marks)

Design an interactive web dashboard where users can:

- **Input Data:**
  - Enter source text (in English or a supported Indic language).
  - Choose the target language for translation.
- **Display Visual Outputs:**
  - **Translation View:** The translated output alongside a visual word-alignment table/heatmap or cross-lingual attention visualization.
  - **Information Extraction (IE) View:**
    - **Named Entities (NER):** Color-coded highlights for Persons, Locations, Organizations, etc.
    - **Relation Extraction:** A visual table or graph mapping relationships between extracted entities (e.g., [Entity A] — (Relation) → [Entity B]).
    - **Events & Time:** Highlighted event triggers paired with extracted temporal expressions/timestamps.
- **Interactive Controls:**
  - Toggle controls to hide/show word alignment or specific IE components.
  - A download button to export the translated text and the extracted IE knowledge base in **JSON** or **CSV**format.

2. MT & Information Extraction Engine (4 Marks)

Implement the backend processing modules using Python and relevant NLP toolkits:

- **Machine Translation (MT) Module:**
  - Implement an MT model (Statistical SMT using Moses/FastAlign OR Neural/Indic MT using frameworks like IndicTrans / HuggingFace Transformers).

    - Perform complete preprocessing (tokenization, normalization, script handling for Indic languages).
- **Information Extraction (IE) Module:**
    - **Named Entity Recognition (NER):** Extract entities from the source or translated text.
    - **Relation Extraction:** Identify and classify semantic relationships between identified entities.
    - **Event & Temporal Extraction:** Extract event occurrences and link them with time expressions.

Task B: Efficiency & System Optimization Plan (2 Marks)

Document a theoretical and empirical plan addressing system optimization and architectural design:

- **Pipeline Strategy:**
    - Compare a **Translate-then-Extract** pipeline vs. a **Direct Cross-Lingual Extraction** model. Discuss trade-offs in error propagation and translation quality.
- **Scalability & Inference Optimization:**
    - Propose model compression methods (e.g., quantization, pruning, or knowledge distillation) to optimize the MT and IE models for low-latency web deployment.

## Deliverables

PART – 1

1. **Source Code:** Well-documented Python code containing the frontend interface, MT backend, and Information Extraction modules.
2. **Setup Guide:** Clear instructions for installing dependencies and running the application locally.
3. **System Report:** A brief technical report explaining system architecture choices, preprocessing steps, toolkits used, and challenges encountered during Indic translation/IE implementation.
4. **Application Screenshots:** Step-by-step screenshots demonstrating the complete end-to-end workflow (Input →Translation & Alignment → IE Extraction → JSON/CSV Download).
5. **Task-B Document:** Submission of Task B as a .pdf document.